

Enabling Modularity in IoT: Running WebAssembly Containerized Workloads on Microcontrollers

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Project details:

- **Project Title:** Enabling Modularity in IoT: Running WebAssembly Containerized Workloads on Microcontrollers
- **Project supervisor:** Professor Stefano Vissicchio

Executive summary:

This project evaluates the feasibility of deploying containerized control workloads on resource-constrained microcontrollers (MCUs) using WebAssembly (Wasm). The scope focuses on decomposing control algorithms into isolated Wasm modules and benchmarking their performance against native C/C++ implementations. The project follows a two-phase approach: characterizing container performance on a single device, and subsequently assessing the scalability of the architecture across an IoT network.

Aims and Objectives:

- Explore the WASM project and its ecosystem
- Review existing literature on WASM containers running on IoT devices
- Run a control workload on a MCU as a container
- Explore containerization strategies on an IoT network
- Experiment with different containerization strategies and workloads

Deliverables:

- Literature review
- Code for running containerized control workload(s) on MCU
- Performance analysis of different containerization strategies
- Final project report

Work Plan:

- **3rd week of September to end of October (4 weeks):**
 - Conduct initial research into Web Assembly project and its ecosystem
 - Explore workloads to run on MCU
- **1st week of November to 1st week of December (5 weeks):**
 - Decide on an initial control workload to run on the MCU
 - Setup tooling to work with WASM
 - Setup MCU to run WASM containers

- Conduct literature review
- **2nd week of December to 2nd week of January (5 weeks):**
 - Depending on control task, determine factors of interest eg power consumption
 - Get the control workload working as a native version and as a single container
 - Measure performance with respect to factors of interest
 - Submit interim report
- **3rd week of January to 1st week of February (3 weeks):**
 - Experiment with different containerization strategies on a single MCU
 - Measure performance with respect to factors of interest
- **Middle of February to middle of March (5 weeks):**
 - Extend the project by running multiple containerized workloads in an IoT network
 - Explore different containerization strategies across this IoT networks
- **Middle of March to middle of April (4 weeks):**
 - Analyse performance results for different containerization strategies
 - Work on final report, detailing methodology, results and recommendations for future work

Hardware resources:

- ESP32 microcontroller
- Equipment to measure performance of ESP32
- Actuators and sensors for the control workload
- The above will be made available from the project supervision team

Relevant work:

- WebAssembly: [WASM](#)
- WebAssembly Micro Runtime: [WAMR](#)
- ESP-IDF: [ESP-IDF](#)
- Skrivankova, K., Handley, M. and Hailes, S. (2025). 20 Years in Life of a Smart Building: A retrospective. [online] arXiv.org. Available at: <https://arxiv.org/abs/2509.06229> [Accessed 9 Nov. 2025]